

Data Architecture (Medallion Architecture)

A Medallion Architecture structures pipelines into three distinct layers: bronze, silver and gold.

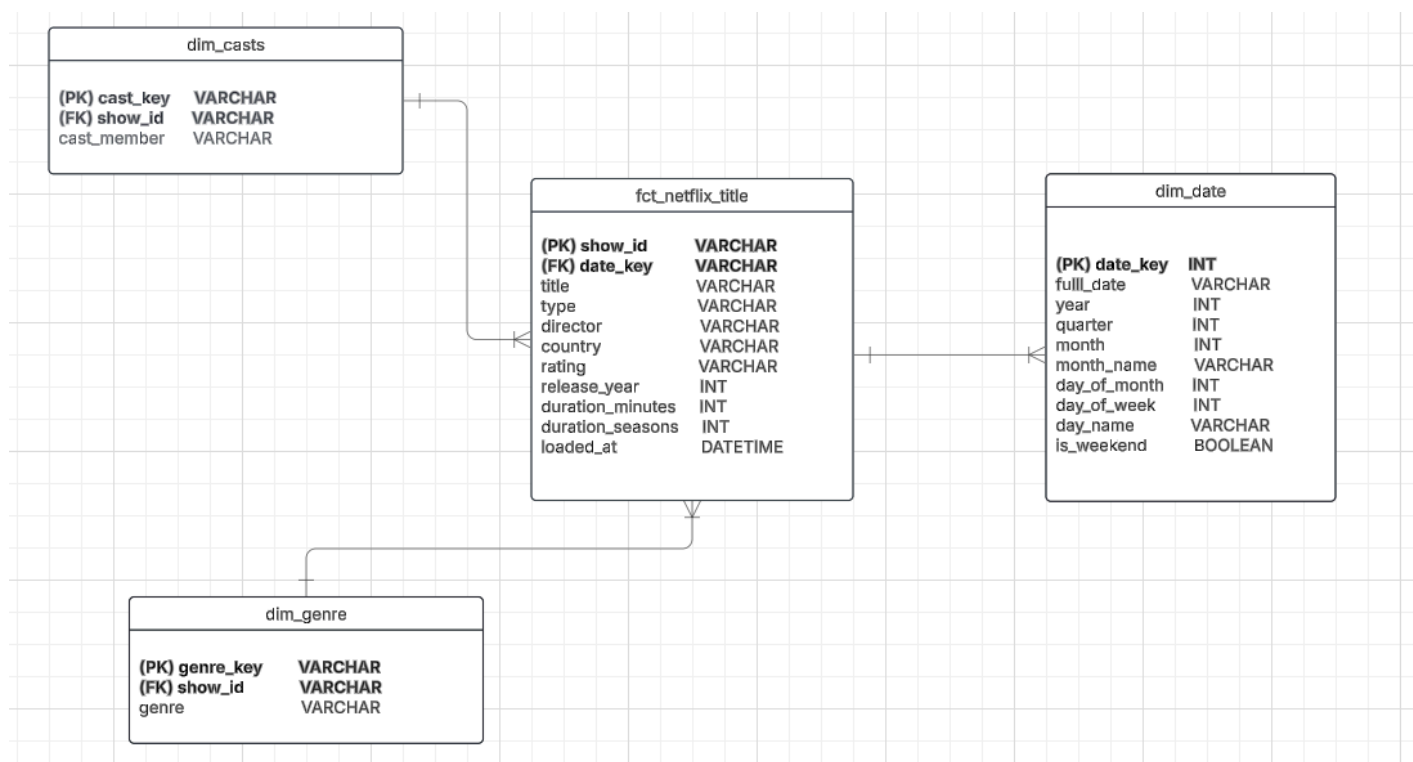
- The bronze layer is the layer where the raw data is stored, acting as a single source of truth.
- The silver layer is the layer where the raw data is then cleansed. Removing duplicates, handling nulls, etc.
- The gold layer is the layer where the fact, dimension tables are stored and aggregations are done such that they are ready for analytical processes.

Why Medallion Architecture?

- Traceability: With the separation of the layers, we can observe and compare the transformations of the different layers. Hence, allowing for easy diagnosis to issues in each layer.
- Decoupled Processing: By having the different layers, I am able to re-process specific layers that I want to. For example, if a business logic changes and I have to amend the gold layer, I do not have to run the bronze or silver layer again. Just re-processing the gold layer would be enough, and it saves time and is more efficient.

Data Modelling (Star Schema)

Star Schema is used in this case to model the data for our gold layer for aggregation.



- From our silver layer, it is separated into fct_netflix_titles, dim_date, and dim_casts tables.
- The tables are not fully normalised as observed. This allows for quicker joins, hence speeding up data aggregations for analysis processes.
- However, this does not allow for optimal scalability as denormalised tables introduce increased complexity as scale increases:
 - Data Duplication
 - Increased risk of inconsistencies

- Complex schema evolutions
- This is not a problem as the dataset that we are dealing with has 12 columns and not a significant amount of data to be handled.

Metabase

Metabase is an open-source business intelligence platform which can create dashboards, visualisations etc, and which allows for self-hosting without any cost.

- With Metabase, we can easily create visualisations based on the fact tables and the data mart that I have created for aggregations.
- Non-Technical Users, would be able to interact with the Metabase UI to create their own visualisations as it can do so without any code.

This allows for many stakeholders to be able to interact with the platform and discussions and analysis can be done by both technical and non-technical members.

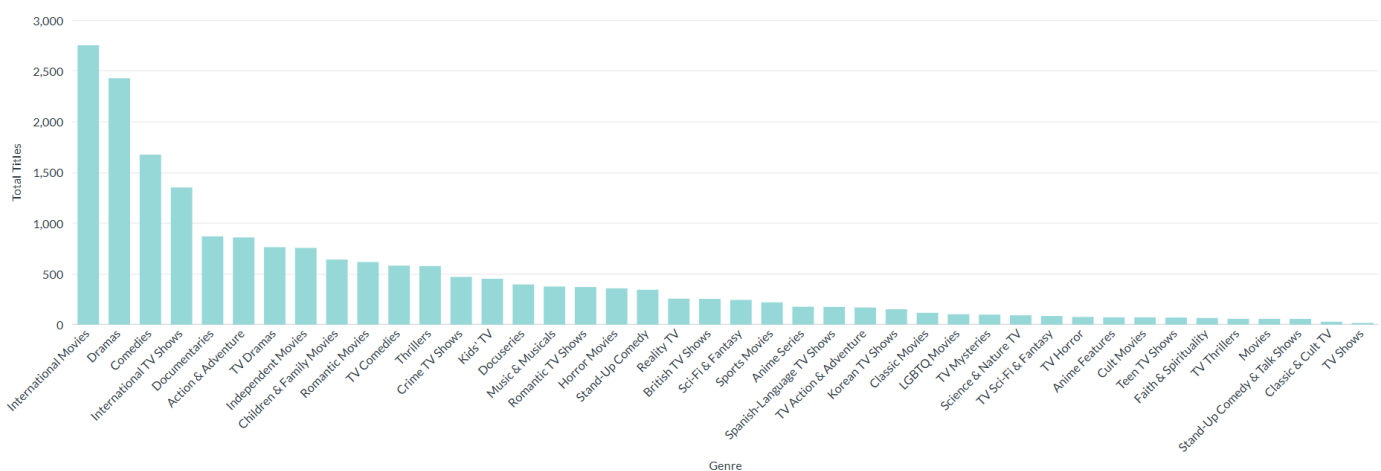
Visualisations

Graph of Number of Titles per Genre

This graph is visualised with the mrt_genre_counts table

Steps to Visualise in Metabase:

- Click into Databases -> <Database_Name> -> gold -> Mrt Genre Counts
- Click into Visualisations -> More Charts -> Bar



Graph of Genre Trends over the Years

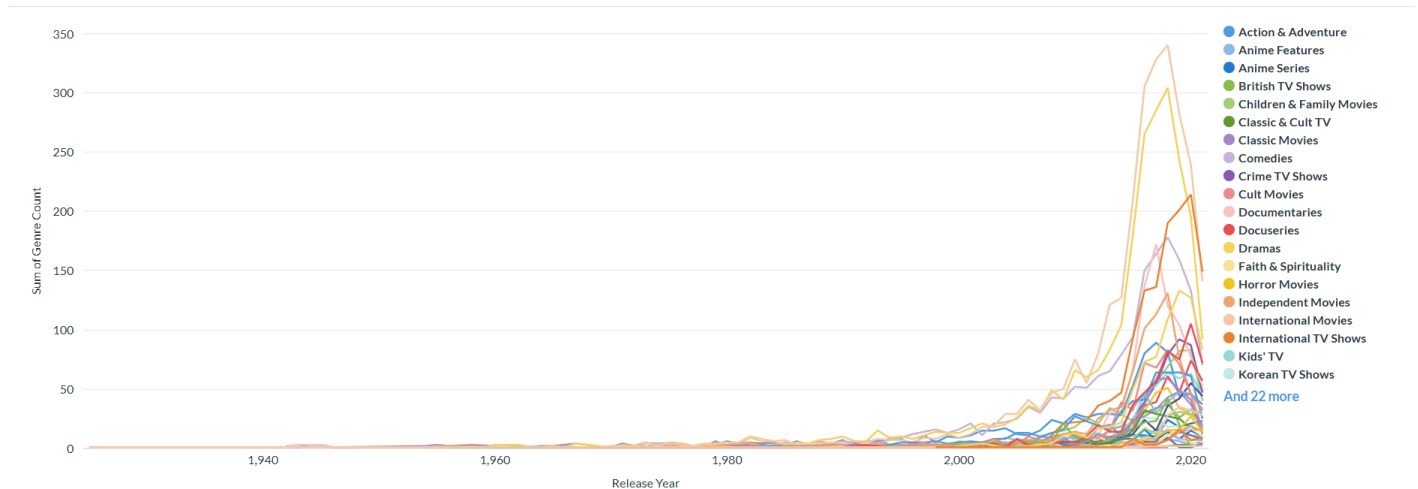
This graph is visualised with the mrt_genre_trends table

Steps to Visualise in Metabase:

- Click into Databases -> <Database_Name> -> gold -> Mrt Genre Trends
- Click into Summarise
- Close the "Count" tab if it is open

- Add a function or metric > Sum of ... > Genre Count
- Click onto the + sign of Genre to add Grouping
- Click onto the + sign of Release Year to add Grouping > Don't Bin
- Visualisation > Line

Now we can hover over each genre to see the genre trends in titles over the years.

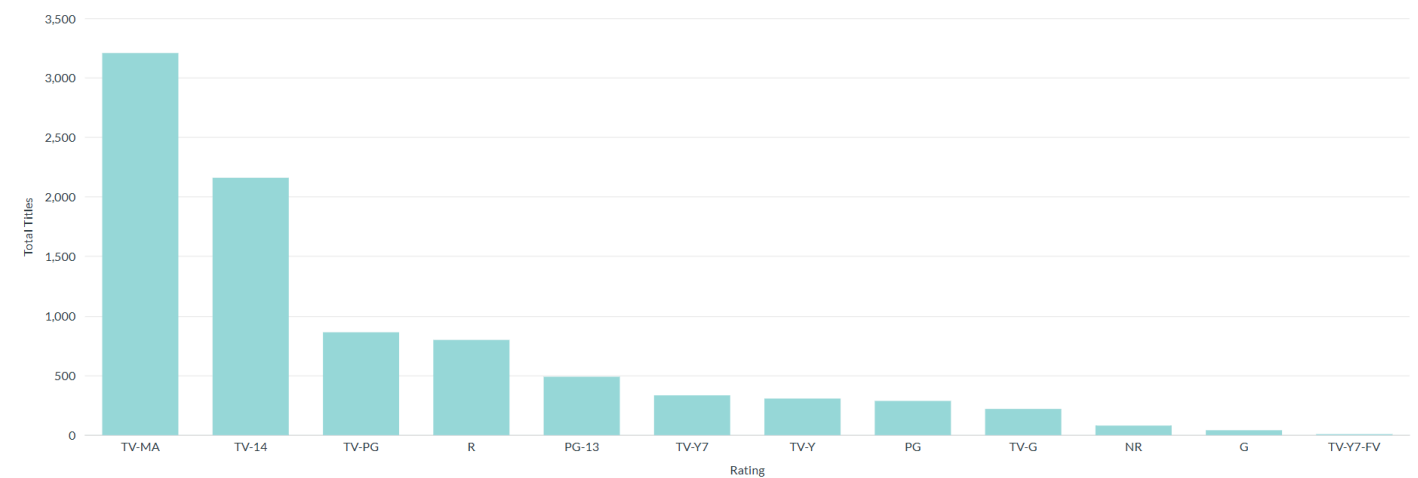


Graph of Number of Titles per Rating

This graph is visualised with the mrt_ratings_counts table

Steps to Visualise in Metabase:

- Click into Databases -> <Database_Name> -> gold -> Mrt Ratings Counts
- Click into Visualisations -> More Charts -> Bar

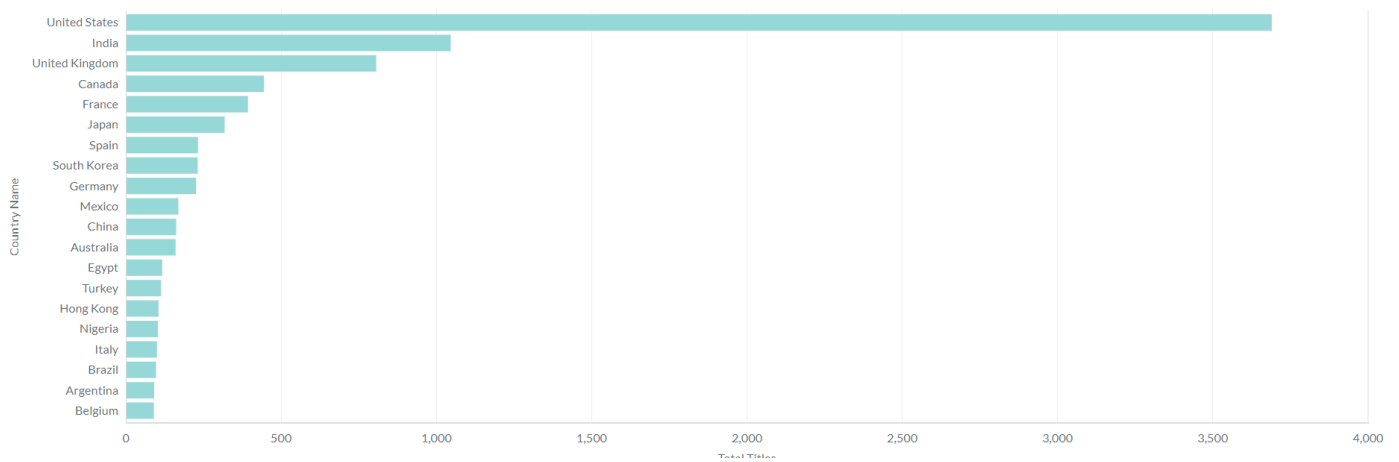


Graph of Number of Titles released per Country

This graph is visualised with the mrt_country_counts table

Steps to Visualise in Metabase:

- Click into Databases -> <Database_Name> -> gold -> Mrt Country Counts
- Click into Visualisations -> More Charts -> Row

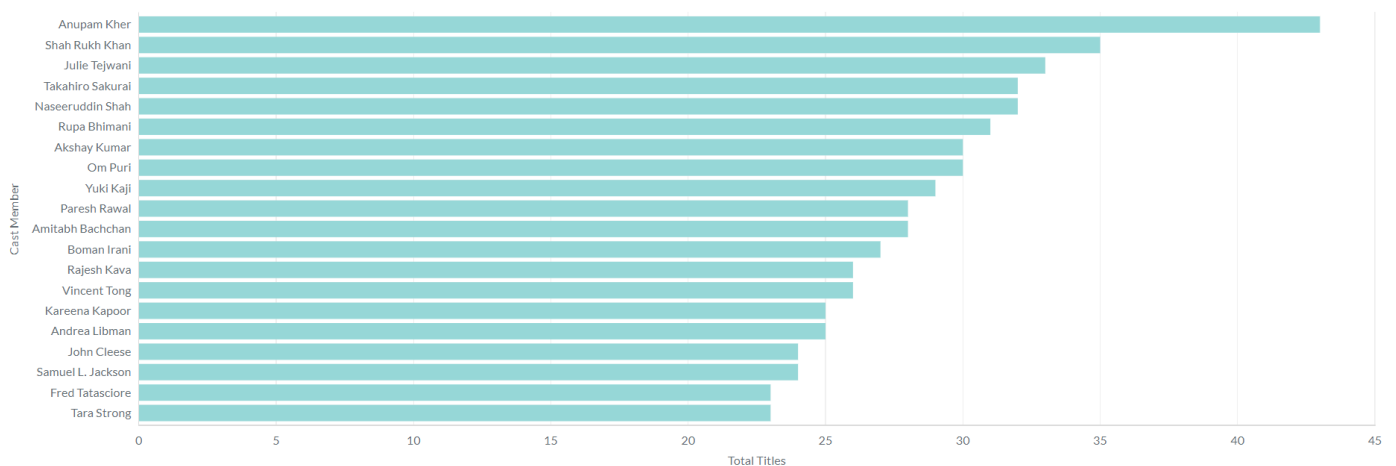


Graph of Number of Titles per Cast Member

This graph is visualised with the mrt_cast_counts table

Steps to Visualise in Metabase:

- Click into Databases -> <Database_Name> -> gold -> Mrt Cast Counts
- Click into Visualisations -> More Charts -> Row



Analysis of the Data

Immediate Tactical Moves and Low-Hanging Fruits

Based on analysis from our genre trends and the frequency of genres appearing in different titles, if we want to compete with Netflix, finding titles with genres which are trending, but however have low counts in Netflix will help us gain a competitive edge.

- For example, the Docuseries genre has an upward trend according to the trend graph, but however has low representation in the titles in Netflix. Hence, we should be pushing to license more titles that is part of the Docuseries genre. This will allow for supply to meet the rising demand for the genre from the customers.

If we would want to compete with them with the trendiest genres and also having high representation in Netflix. We would need to have equal or more titles that have those genres. Furthermore, we might need to offer other benefits to customers to have the competitive edge to pull them away from Netflix

Furthermore, if we look at the the ratings distribution for the titles in Netflix, we can see there are indeed quite a few genres which are in the minority but still very important for a diverse audience base.

- If we look at the ratings, there is an overwhelming skew towards content that are rated mature and not suitable for younger audiences. Hence, by investing in licensing more titles which are overall family friendly, we can market

ourselves as a platform that is more suited for families to enjoy together.

Many parents who has buying power would subscribe to our services as a way to spend time with their family, providing entertainment to their family, while also ensuring that the younger audiences would not easily be influenced by graphic content that may be in the platform. We would also be avoiding Netflix in competing with capturing audiences who enjoy more of those mature content, allowing us to not have such fierce competition.

Mid-Term growth levers, scaling strategies, and content / product investments

Looking at the graph of countries that produce content that are being streamed by Netflix, we can see an overwhelming number of them in United States, followed by India, then United Kingdom. We can see there are many countries which are under-represented in the titles that are in Netflix. There is an opportunity for us to invest in producing content at the countries which are being under-represented.

- Producing exclusive content in those countries, we can involve the local community in the project. This can help with increased representation which can lead to the pull of customers from those countries to our platform to watch the exclusive content.

Since we are focusing on markets that are not heavily in the Netflix ecosystem, we have the chance to create a new customer base while also increasing it at the same time.

Secondly, we can commission actors that frequently appear in Netflix's titles for our own projects. We can identify them based on the how many titles a cast member has appeared in with the graph.

- By approaching these actors with original production, we can pull already established talent in this space from Netflix's contents. At the same time, we can also bring the attention of the fans who religiously follow that actor to our platform, and they would subscribe to our services to watch their production.

A defining, foundational strategy that builds a permanent competitive moat to differentiate from Netflix.

From the actors being commissioned, we can approach them to create a community centered around the actor and their work. This allows for a mutually beneficial relationship where we create an active fan network where the fans involved in that actors community can interact, and watch exclusive content.

- Fans will actively subscribe to our services in order to keep up with the content of the actor and its community.

We are selling a place of belonging, sharing and expression for the fans who loyally follow the actors. This allows for us to not directly compete with Netflix which are just focused on selling subscription services to their customers.

Not only that, we are able to build a brand identity which can shape customer impression of our company. By pivoting away from mature content, and empowering communities under-represented in TV series and movies, we can build a brand identity of family friendly and diverse platform that stands in direct contract to Netflix's more mature, Western dominated catalog.

- Families who associate with our platform as being safe, inclusive and culturally diverse would actively advocate for our product. Word-of-mouth within their own communities, can be one of the most effective way of customer acquisition channels available.
- Furthermore, schools, cultural platforms can both be the consumer and contributor to our company based on this brand identity. They can be partners with us, creating educational content on the under-represented communities around us, and also be subscribed to our product.